Reasoning test

For admission to Class 10

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 9 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 9.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, on these topics.

Maths

1. Number patterns and series
2. Primes, factors, HCF and LCM
3. Fractions
4. Perimeter and area
5. Simple equations
6. Speed, distance and time
7. Averages
8. Ratio and proportion
9. Percentages, profit and loss, interest
10. Squares and cubes
11. Probability

English

1. Vocabulary
2. Grammar
3. Sentence completion and ordering
4. Reading comprehension

Science

1. Evaporation and condensation
2. Magnets
3. Light, shadows and mirrors
4. Acids, bases and indicators
5. Physical and chemical changes
6. Heat
7. Electric circuits
8. Respiration
9. Photosynthesis
10. Motion and speed
11. Food chains
12. Floating, sinking and density
13. Force and pressure
14. Sound
15. Mass and weight
16. Force and laws of motion
17. Change of state
18. Elements, compounds and mixtures
19. Energy

Reasoning

1. Clocks and calendars
2. Number codes and puzzles
3. Everyday units
4. Word problems
5. Time and work
6. Problems on ages
7. Mixtures
8. Venn diagrams
9. Analogies, odd one out, series and codes
10. Logical deduction
11. Reading an experiment
12. Applying a principle to a new situation

Class 9 syllabus questions

Maths

1. Orienting Yourself: The Use of Coordinates
2. Introduction to Linear Polynomials
3. The World of Numbers
4. Exploring Algebraic Identities
5. I'm Up and Down, and Round and Round

Science

1. Cell – The Building Block of Life
2. Tissues in Action
3. Describing Motion Around Us
4. Exploring Mixtures and their Separation
5. How Forces Affect Motion

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   What is the angle between the hands of a clock at exactly 3 o'clock?
   A 45° **B** 60°
   C 75° **D** 90°

2. ENGLISH · BASICS
   The general's plan was simple but ingenious. As used here, "ingenious" most nearly means
   A dangerous **B** clever
   C costly **D** careless

3. SCIENCE · BASICS
   As a ball falls freely towards the ground, which energy change takes place?
   A potential energy to kinetic energy **B** kinetic energy to potential energy
   C kinetic energy to chemical energy **D** heat energy to potential energy

4. MATHS · CLASS 9
   On which axis does the point (0, −4) lie?
   A the x-axis **B** the y-axis
   C neither axis **D** both axes

5. ENGLISH · BASICS
   Choose the correct words: The bundle of letters ___ on the table.
   A were lying **B** are lying
   C was lying **D** have lain

6. SCIENCE · CLASS 9
   A train covers 150 km in 2.5 hours. What is its average speed, in km/h?
   Type-in answer: _____

7. MATHS · BASICS
   The price of a bag rises from ₹250 to ₹300. What is the percentage increase?
   A 15% **B** 20%
   C 25% **D** 50%

8. ENGLISH · BASICS
   Ear is to hearing as eye is to ___
   A tears **B** face
   C colour **D** sight

9. SCIENCE · BASICS
   An astronaut has a mass of 60 kg on Earth. What is her mass on the Moon?
   A 0 kg **B** 10 kg
   C 60 kg **D** 360 kg

10. MATHS · CLASS 9
    If p(x) = 4x + 7, what is the value of p(3)?
    Type-in answer: _____

11. ENGLISH · BASICS
After the heavy rain, the hill path became treacherous. As used here, "treacherous" most nearly means

A dangerous
B narrow
C muddy
D steep

12. SCIENCE · CLASS 9
Which part of a plant cell contains chlorophyll and makes food?

A Nucleus
B Chloroplast
C Vacuole
D Cell wall

13. MATHS · BASICS
A car uses 8 litres of petrol to travel 96 km. How many litres will it need to travel 156 km?

Type-in answer: _____

14. ENGLISH · BASICS
Choose the correct word: The train arrived ___ the station on time.

A in
B on
C at
D to

15. SCIENCE · BASICS
An ant bite injects an acid into the skin. Which of these would best relieve the sting?

A a few drops of vinegar
B baking soda paste
C fresh lemon juice
D plain salt water

16. MATHS · CLASS 9
What do you get when you expand $(a - 3)(a + 3)$?

A $a^2 - 9$
B $a^2 + 9$
C $a^2 - 6a - 9$
D $a^2 - 3$

17. ENGLISH · BASICS
Author is to book as composer is to ___

A orchestra
B piano
C stage
D music

18. SCIENCE · CLASS 9
Which method recovers the salt from salt water?

A Filtration
B Decantation
C Evaporation
D Chromatography

19. MATHS · BASICS
₹5,000 is deposited at 10% interest per year, compounded yearly. What will the amount be after 2 years?

A ₹5,500
B ₹6,000
C ₹6,050
D ₹6,500

20. ENGLISH · BASICS
"The bus was late, the rain soaked us and the shop was shut. What a perfect day!" The speaker's tone is

A sarcastic
B sincere
C hopeful
D indifferent

21. SCIENCE · BASICS
40 g of sugar is dissolved in water to make 500 ml of solution. What is the concentration, in grams per litre?

Type-in answer: _____

22. MATHS · CLASS 9

How far is the point (6, 8) from the origin?

Type-in answer: _____

23. ENGLISH · BASICS

What comes next in the series? B, E, H, K, ___

A M
B N
C O
D L

24. SCIENCE · CLASS 9

A net force of 12 N acts on a 4 kg trolley. What is its acceleration?

A 0.33 m/s^2
B 3 m/s^2
C 16 m/s^2
D 48 m/s^2

25. MATHS · BASICS

In a class of 50 students, 30 play cricket, 25 play football and 8 play neither game. How many play both?

A 8
B 13
C 17
D 22

26. ENGLISH · BASICS

"New traffic lights were installed in May and road accidents fell in June, so the lights prevent accidents." Which fact, if true, most weakens this argument?

A The lights cost the town a lot
B Some drivers dislike waiting at lights
C Schools closed in June, cutting traffic
D Accidents also happen late at night

27. SCIENCE · BASICS

A burn from steam at 100 °C is usually worse than one from boiling water at 100 °C. This is because steam

A is at a higher temperature than water
B moves faster and hits the skin harder
C carries germs that the water does not
D gives out extra heat as it condenses

28. MATHS · CLASS 9

AB is a diameter of a circle and C is a point on the circle. If ∠CAB = 35°, what is ∠CBA?

A 35°
B 45°
C 55°
D 90°

29. ENGLISH · BASICS

No soldier is careless. Some cadets are careless. Which statement must be true?

A No cadet is a soldier
B Some cadets are not soldiers
C All soldiers are cadets
D Some soldiers are careless

30. SCIENCE · CLASS 9

A cyclist moving at 4 m/s speeds up uniformly at 0.5 m/s^2 for 6 seconds. What is the final velocity, in m/s?

Type-in answer: _____

Answers

1	**D. 90°**	The clock face is 360°, and each hour mark is 30° apart. From 12 to 3 is three hour marks, so the angle is 3 × 30° = 90°.
2	**B. clever**	Ingenious means clever and inventive. "Simple but" signals a quality that is praised, which rules out careless and dangerous.
3	**A. potential energy to kinetic energy**	As the ball falls it loses height, so it loses potential energy, and it speeds up, so it gains kinetic energy.
4	**B. the y-axis**	Its x-coordinate is 0, so the point is on the y-axis, 4 units below the origin. Points on the x-axis have y-coordinate 0 instead.
5	**C. was lying**	The subject is "the bundle", which is singular; "of letters" only describes it. So the verb is "was lying".
6	**60**	Average speed is total distance divided by total time: 150 km ÷ 2.5 h = 60 km/h. This is a type-in question: the answer is typed as a number.
7	**B. 20%**	The rise is ₹50, and the percentage is worked out on the original price: 50 ÷ 250 = 0.2, which is 20%.
8	**D. sight**	The pair links an organ to the sense it provides. The ear gives hearing and the eye gives sight.
9	**C. 60 kg**	Mass is the amount of matter and does not change from place to place. Her weight on the Moon is about one sixth, but her mass stays 60 kg.
10	**19**	Put x = 3 into the polynomial: 4 × 3 + 7 = 12 + 7 = 19. This is a type-in question: the answer is typed as a number.
11	**A. dangerous**	Treacherous means full of hidden danger. The rain may well have made the path muddy, but that is a cause, not the meaning of the word.
12	**B. Chloroplast**	Chloroplasts hold the green pigment chlorophyll and carry out photosynthesis. The nucleus controls the cell, the vacuole stores, the cell wall supports.
13	**13**	The car runs 96 ÷ 8 = 12 km on each litre, so 156 km needs 156 ÷ 12 = 13 litres. This is a type-in question.
14	**C. at**	We arrive "at" a place such as a station, and "in" a city or country. "Arrived to" is a common slip, but it is not standard English.
15	**B. baking soda paste**	An acid is neutralised by a base. Baking soda is a mild base; vinegar and lemon juice are acids and would not help.
16	**A. $a^2 - 9$**	The identity $(x - y)(x + y) = x^2 - y^2$ gives $a^2 - 3^2 = a^2 - 9$. The middle terms $-3a$ and $+3a$ cancel, so there is no a-term.
17	**D. music**	An author creates a book; a composer creates music. An orchestra and a piano perform it, and a stage is where it is performed.
18	**C. Evaporation**	Salt is dissolved, so it passes through filter paper and cannot be poured off. Evaporating the water leaves the salt behind.

19	**C. ₹6,050**	After year 1: 5,000 + 500 = ₹5,500. Year 2 earns 10% of 5,500, which is ₹550, giving ₹6,050. ₹6,000 is simple interest.
20	**A. sarcastic**	Everything described went wrong, so calling it a perfect day means the opposite of what is said. That is sarcasm.
21	**80**	500 ml is half a litre. 40 g in half a litre is the same as 80 g in a full litre. This is a type-in question.
22	**10**	The distance from the origin is $\sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$. This is a type-in question: the answer is typed as a number.
23	**B. N**	Each letter is three places after the one before: B (+3) E (+3) H (+3) K (+3) N. M and L come from counting two or one place instead of three.
24	**B. 3 m/s²**	From F = ma, a = F ÷ m = 12 N ÷ 4 kg = 3 m/s². 48 multiplies instead of dividing, and 0.33 divides the wrong way round.
25	**B. 13**	50 − 8 = 42 students play at least one game. 30 + 25 = 55 counts the players of both games twice, so 55 − 42 = 13 play both.
26	**C. Schools closed in June, cutting traffic**	If traffic itself fell in June, that alone could explain fewer accidents, so the lights may not be the cause. The others do not touch the cause.
27	**D. gives out extra heat as it condenses**	Both are at 100 °C, but steam releases a large amount of extra heat (latent heat) when it turns back into water on the skin.
28	**C. 55°**	The angle in a semicircle is 90°, so ∠ACB = 90°. The angles of triangle ABC add to 180°, so ∠CBA = 180° − 90° − 35° = 55°.
29	**B. Some cadets are not soldiers**	The careless cadets cannot be soldiers, so at least some cadets are not soldiers. Nothing rules out other cadets being soldiers, so "no cadet is a soldier" goes too far.
30	**7**	Using v = u + at: v = 4 + 0.5 × 6 = 4 + 3 = 7 m/s. This is a type-in question: the answer is typed as a number.